- CPSC 375: High-Performance Computing

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Spring 2026

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- The Cluster Project

Homework 23

NOTE: You are not required to hand in the following exercises, but you are strongly encouraged to complete them to strengthen your understanding of the concepts covered in class.

1. Consider a simple block allocation of n data items to p processes in which the first $p - 1$ processes get $\lceil n/p \rceil$ items each and the last process gets what is left over.

   A. Find values for n and p where the last process does not get any elements.
   B. Find values for n and p where $\lfloor p/2 \rfloor$ processes do not get any values. Assume $p > 1$.

2. We discussed two block data decomposition strategies that assign n elements to p processes such that each process is assigned either $\lfloor n/p \rfloor$ or $\lceil n/p \rceil$ elements. For each pair of values n and p, use a table or an illustration to show how these two schemes would assign array elements to processes:

   A. $n = 15$ and $p = 4$
   B. $n = 15$ and $p = 6$
   C. $n = 16$ and $p = 5$
   D. $n = 18$ and $p = 4$
   E. $n = 20$ and $p = 6$
   F. $n = 23$ and $p = 7$

- **Welcome: Sean**
  - LogOut

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